

Learning syntax without semantics: Disentangled tiny language models

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Abstract

Language models acquire syntax and world knowledge together, entangling the two in ways that limit efficiency and controllability. We show that syntax can be learned while suppressing semantic plausibility and world-knowledge cues, yielding more efficient and controllable models. We train tiny LMs on grammatical nonsense—syntactically well-formed text with semantic content ablated via constrained relexicalization (SAMBAL). Models trained on this data perform comparably to standard pretraining on syntactic benchmarks (BLiMP, SyntaxGym) while scoring at chance on world knowledge probes (EWoK). On targeted grammar-plausibility conflict probes, content-neutral models prefer grammaticality where standard models prefer plausibility, and their representations show more syntactic vs. lexical alignment. On efficiency, disentanglement yields substantial sample and parameter gains in low-resource regimes: a 5M-parameter model matches a 30M-parameter baseline at the same data budget. On controllability, content-neutral models adapt rapidly to a new domain with minimal exposure, suggesting the feasibility of modular post-hoc knowledge specialization.

1. Introduction

Humans acquire core language abilities from vastly less linguistic exposure—often estimated at $\lesssim 10^8$ words—than modern language models, which are typically trained on 10^{11} – 10^{12} + tokens (Charpentier et al., 2025; Hu et al., 2024; Hoffmann et al., 2022). One key difference is that LMs are trained to serve multiple roles, as statistical models of linguistic form, advanced reasoners, and repositories of world knowledge and semantic plausibility (Mahowald

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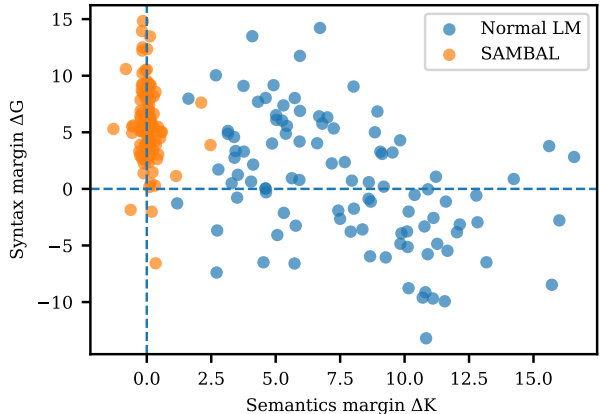


Figure 1. Syntax margin vs. semantic-plausibility margin on minimal sentence pairs. Models trained on grammatical nonsense (SAMBAL, orange) show a preference for syntactically-correct sentences, with positive logit margin $\Delta G = \log P(\text{grammatical}) - \log P(\text{ungrammatical})$, but no preference for semantically-plausible sentences, with logit margin $\Delta K = \log P(\text{plausible}) - \log P(\text{implausible})$ near zero, unlike a normally trained LM (§4.3).

et al., 2024; Bender & Koller, 2020). Whether syntax can be learned independently of meaning is a central question in linguistics, spanning arguments for the autonomy of syntax, poverty-of-the-stimulus debates, and evidence that learners can extract structure from distributional regularities (Chomsky, 1957; 1965; Pullum & Scholz, 2002; Saffran et al., 1996). This raises a concrete question for language modeling: can syntax be learned more efficiently if we actively suppress semantic and world-knowledge cues?

If syntax can be isolated from knowledge, two practical benefits follow. First, *efficiency*: if syntax requires less data than syntax-plus-knowledge, then isolating syntax learning could yield sample and parameter gains—relevant both for low-resource settings and for understanding what inductive biases support language learning (Kaplan et al., 2020; Hoffmann et al., 2022; Warstadt et al., 2023; Hu et al., 2024; Charpentier et al., 2025). Second, *controllability*: knowledge learned parametrically is notoriously hard to update, edit, or specialize for new domains (Meng et al., 2022; 2023). Prior work addresses these problems post hoc—via steering, editing, and unlearning (Liu et al., 2024; Zhao et al., 2024;

Wang et al., 2024; Yao et al., 2024). Our work suggests a different approach: suppress knowledge acquisition during pretraining, yielding a syntax-competent base onto which domain knowledge can later be added modularly.

Our approach. We hypothesize that a tiny LM can learn *syntactic competence* while suppressing *semantic plausibility and world-knowledge cues*. To test this, we train on *grammatical nonsense*—text that is syntactically well-formed yet semantically anomalous, in the spirit of Chomsky’s classic “colorless green ideas” (Chomsky, 1957; Gulordava et al., 2018). Our pipeline, SAMBAL¹, transforms natural text into grammatical nonsense through constrained relexicalization: SAMBAL preserves morphosyntax, agreement, and complement structure (c-selection) while ablating plausibility priors (s-selection) as well as factual and world knowledge. For example:

Original: *Adopting instead of shopping reduces demand for commercially bred pets and gives loving homes to those already waiting.*

SAMBAL: *Eating instead of learning gives probability for occasionally balanced men and creates architectural tears to those now living.*

We pretrain tiny LMs on SAMBAL-transformed corpora of 10M tokens or less, and evaluate them along several axes.

Contributions and findings. (1) **Syntactic competence.** SAMBAL matches standard pretraining on the grammar benchmarks BLiMP (Warstadt et al., 2020) (79.3% vs. 79.2% acc.) and most SyntaxGym (Gauthier et al., 2020) suites (79.8% vs. 78.6% acc.), with the notable exception of reflexive SyntaxGym suites. (2) **Semantic suppression.** SAMBAL falls to chance on the world-knowledge probe EWoK (Ivanova et al., 2025) (50.8% vs. 54.3%) and shows near-chance semantic-plausibility preference on a custom swap probe (55% vs. 100%). (3) **Reduced entanglement.** On a grammar-plausibility conflict probe, SAMBAL prefers grammaticality where the baseline model prefers plausibility; representations show higher alignment to syntax relative to semantics. (4) **Efficiency gains.** SAMBAL is more efficient across a range of very small data and parameter budgets; with 1M words, a 5M-parameter SAMBAL model matches a 30M-parameter baseline. (5) **Modular specialization.** SAMBAL models (by design) have high perplexity on ordinary text, but they adapt rapidly via LoRA (Hu et al., 2022) to small domain-specific corpora—both 580k words of *Lord of the Rings* (Tolkien, 1994) and a similarly-sized set of PubMed abstracts (NLM, 2024)—regimes where training a competent model from scratch would be infeasible.

¹<https://github.com/ezrawinston/sambal>

2. Background and related work

Our approach is motivated by the hypothesis that syntax and world knowledge, while entangled in standard pretraining, may be realized as distinct mechanisms that can largely be teased apart. This connects with several established research threads: (i) the form–meaning distinction in language, (ii) separability of syntax in model representations, (iii) child-scale / tiny-LM efficiency, (iv) shaping the pretraining signal toward structure, and (v) modular, updatable knowledge.

Syntax, semantics, and separability. The distinction between linguistic form and communicative meaning is a foundational debate in cognitive science. Classically, the *autonomy of syntax* hypothesis posits that the combinatorial rules of language operate independently of concepts or communicative intent (Chomsky, 1957; 1965)—as in “Colorless green ideas sleep furiously,” grammatically well-formed despite its semantic vacuity. This separation of form from meaning has been argued to persist in language models, which can master linguistic form without correspondingly grounded meaning (Bender & Koller, 2020). Mahowald et al. (2024) formalize this form–meaning distinction as *formal linguistic competence*—knowledge of linguistic rules and patterns—versus *functional linguistic competence*—non-linguistic cognitive abilities such as formal reasoning and world knowledge.

Separability of syntax in model representations. Whether trained models represent syntax independently of meaning has been studied largely through their internal representations. Structural probes recover syntactic dependency trees linearly from contextual embeddings, indicating a distinct syntactic code (Hewitt & Manning, 2019). This code appears genuinely structural rather than a reflection of semantic predictability: probe accuracy is governed by properties such as dependency length and depth and is largely unaffected by a word’s surprisal (Diego-Simón et al., 2025). Consistent with separability, syntactic structure remains decodable even when lexical meaning is removed: under grammaticality-controlled nonce (*Jabberwocky*) relexicalization, decoding drops only modestly and most of the syntactic signal is preserved (Arps et al., 2024), though earlier pseudoword studies reported larger drops (Maudslay & Cotterell, 2021). A recurring caveat in this literature is grammaticality itself: relexicalizing on part-of-speech alone frequently breaks argument structure, yielding sentences rated grammatical only ~38% of the time versus ~87% when valency is preserved (Gulordava et al., 2018; Arps et al., 2024), which motivates the careful design of our relexicalization pipeline. Overall, these studies probe whether syntax is *decodable* from a fixed model after meaning is removed at evaluation time; we instead remove meaning throughout training and ask whether syntax is still *learnable*.

Tiny LMs and child-scale pretraining. Whether syntax can be acquired from the limited input available to children—or instead requires innate structural bias—is long debated: poverty-of-the-stimulus arguments hold that the input underdetermines the grammar (Pullum & Scholz, 2002), whereas infants demonstrably extract structural regularities from distributional cues alone (Saffran et al., 1996). Language models let us probe this at comparable data scales (< 100M words), driving efforts such as the *BabyLM Challenge* (Warstadt et al., 2023; Hu et al., 2024). Models with 10M–100M parameters can achieve strong BLiMP performance, especially with sample-efficient objectives (e.g., hybrid causal/masked objectives) and strong optimization or distillation (Warstadt et al., 2023; Charpentier & Samuel, 2024; Hu et al., 2024; Timiryasov & Tastet, 2023). These findings have inspired *Fundamental Language Models* (FLMs)—compact models trained for linguistic proficiency while minimizing stored world knowledge (Collado-Montañez et al., 2025). We build on these architectures and objectives (GPT-BERT), while explicitly enforcing suppression of world knowledge during training.

Shaping the pretraining signal for syntax. A complementary line of work improves grammatical learning by shaping the pretraining *signal* or objective rather than the model, through explicit structural supervision or auxiliary syntactic objectives (Wilcox et al., 2020; Lindemann et al., 2024). Most relevant to us are interventions on the training *data* itself: pretraining on child-directed speech imparts a stronger hierarchical-syntax inductive bias than standard corpora (Mueller & Linzen, 2023), and pre-pretraining on a formal language whose hierarchical structure mirrors natural language transfers a structural bias that improves syntactic generalization and natural-language sample-efficiency (Hu et al., 2025). These approaches motivate our data-centric view, but our intervention runs in the opposite direction: where they *add* structural bias to improve grammar acquisition, SAMBAL *removes* semantic content to isolate syntax from meaning.

Controllability and modular knowledge. A growing literature aims to make model behavior easier to update and specialize without rewriting a base model. Parameter-efficient adaptation (e.g., LoRA-style adapters) enables low-cost, composable specialization (Hu et al., 2022), while post-hoc interventions such as steering, model editing, and unlearning can modify or remove targeted behaviors after pretraining (Meng et al., 2023; Zhao et al., 2024). Our approach is more direct, ablating the pretraining data so plausibility and world knowledge are largely absent during pretraining. Our small-domain adaptation studies on *Lord of the Rings* and PubMed are a small step toward modularity: a content-neutral syntactic base should admit rapid, low-data domain adaptation by leveraging its syntactic competence.

Table 1. What SAMBAL preserves vs. ablates. Borderline items are meaning-adjacent but are retained because they directly affect acceptability.

Preserve: Closed-class items; NPIs and NPI licensors; protected MWEs; morphosyntactic features; agreement; complement types / subcategorization (c-selection); raising/control/ECM signatures

Borderline: Grammaticized features that look semantic but affect well-formedness (e.g., gender and humanness for agreement/binding; noun mass/count class when determiners depend on them)

Ablate: *Semantic plausibility:* selectional preferences (s-selection); typical event/role co-occurrence patterns. *World knowledge:* named entities; topical content; factual/encyclopedic associations

3. SAMBAL: grammatical nonsense via constrained relexicalization

3.1. Goals and terminology

SAMBAL targets *grammatical nonsense*: outputs that remain syntactically well-formed while suppressing cues that support world knowledge or semantic plausibility. By *syntax*, we mean the formal constraints that govern well-formedness (categories/features, dependencies, agreement, binding, complement structure); by *semantics* we mean the meaning-bearing content (open-class lexical meaning, compositional interpretation, and background knowledge that makes an utterance informative or plausible). Within semantics we further distinguish *world knowledge*—encyclopedic/factual and basic conceptual knowledge—from *semantic plausibility*—the lower-level selectional and event-coherence cues that make an utterance interpretable; the SAMBAL ablation targets both.²

What we preserve vs. ablate. Which aspects to preserve or ablate is nuanced; we defer details to Appendix K and summarize the default partition in Table 1. We first preserve core morphosyntactic and licensing constraints (e.g., agreement morphology: *this dog runs* vs. *this dog run*); closed-class items such as determiners/auxiliaries/complementizers (e.g., *a, the, has, are, that, whether*); and binding/licensing configurations such as reflexives and negative polarity items (NPIs) acceptable only in certain licensing environments (*I didn’t see any dogs* vs. *I saw any dogs*).

When relexicalizing open-class items (content nouns/verbs/adjectives/adverbs), the principle is: *preserve*

²We often use *grammar* and *syntax* interchangeably. We use *semantics* in the colloquial sense, excluding the structural, compositional phenomena that formal semantics treats and that SAMBAL in fact preserves (see Appendix K and §4.3).

Algorithm 1 Constrained relexicalization (SAMBAL) for a sentence $x = [x_1, \dots, x_n]$

- 1: Parse sentence x to obtain tokens $[t_1, \dots, t_n]$, POS/morph, dependencies.
- 2: Freeze protected tokens/spans (e.g., closed class and MWEs).
- 3: **for** each replaceable token t_i **do**
- 4: Compute syntactic context key $k(t_i)$ (POS, morph, dep role, head features, frame features).
- 5: Retrieve candidate lemma set $C(k(t_i))$ from bucketed lexicon (and optional to-inf/control filters).
- 6: Optionally intersect $C_i = C(k(t_i))$ with context-conditioned candidate set from corpus stats.
- 7: Sample replacement lemma $\ell_i \sim \text{Sampler}(C_i)$.
- 8: Realize surface form $s(\ell_i)$ via morphological inflection under original morph features.
- 9: **end for**
- 10: Optionally, round-trip parse and reject-resample up to N times if invariants fail.
- 11: Output augmented sentence $x' = [s(\ell_1), \dots, s(\ell_n)]$.

c-selection, *ablate s-selection*. *C-selection* refers to lexically governed requirements on complement *category* and realization (e.g., whether a verb takes an NP/PP/CP/to-infinitive: *depend on [PP]*, *believe [CP]*, *want [to VP]*; and raising/control/ECM signatures, e.g., *seem [to VP]* vs. *persuade NP [to VP]* vs. *believe NP [to VP]*). *S-selection* refers to the semantic type of arguments and events (selectional restrictions and plausibility, e.g., *devour* takes an edible object; *repair*, an artifact), which we suppress.

Borderline cases. Some distinctions (notably gender/humanness/animacy, and polarity sensitivity via NPIs/licensors) are meaning-adjacent yet tightly coupled to acceptability, so we preserve them by default; Appendix K motivates this with examples.

3.2. Constrained relexicalization pipeline

SAMBAL replaces eligible open-class tokens with new lemmas sampled from constraint-matched pools, then realizes surface forms to match the original morphosyntactic features (Algorithm 1). We highlight key aspects here and defer details to Appendix A.

Replacing from syntactically-constrained candidate sets.

For each replaceable token, we compute a *syntactic context key* from the parse (POS, morphosyntactic features, dependency role, and where available subcategorization cues) and retrieve candidates from a *bucketed lexicon* built from external resources (e.g., WordNet (Miller, 1995)/VerbNet (Schuler, 2005)). These restrict replacements to lemmas with compatible syntactic behavior, preserving *c-selection*

(e.g., PP-selecting verbs for *depend on [PP]*, CP-taking verbs for *believe [CP]*, to-infinitival verbs for *want [to VP]*), while leaving semantic fit unconstrained.

Lexicon coverage can be sparse and idiosyncratic, so recorded frames may reflect rare uses rather than typical distribution (e.g., *smile* is almost always intransitive, yet can appear with object-like complements: *smiled a smile* vs. *smiled a book*). Moreover, for certain constructions (e.g., NPI licensors or fine-grained to-infinitival buckets), existing lexicons are unavailable or incomplete; we build targeted lexica ourselves (Appendix A).

Context-conditioned sampling. Naïve swapping from external lexicons can yield ungrammatical outputs (e.g., *smiled a book*) even under strong syntactic constraints, a tradeoff: lexicon/frame matching that is *permissive* admits noisy ill-formed outputs, while *strict* matching means many contexts have *no* candidates. We mitigate this using corpus statistics from the original training corpus keyed on the context features above: a *back-off chain* (fine to coarse) backs off until the candidate pool exceeds a minimum size, then samples in proportion to observed frequency.

Round-trip validation and reject-resample. We optionally reparse the augmented sentence and reject/resample if key invariants fail (e.g., POS/morph mismatches, or violations of protected configurations such as NPI licensing), guarding against noisy parsing and realization errors, without manual rules for every construction.

4. Experiments

4.1. Setup

All of our experiments are based on the 10M-word `baby-cosmo-fine` corpus and small GPT-BERT hybrid model (30M params), the winning combination from the 2024 BabyLM competition (Hu et al., 2024). GPT-BERT (Charpentier & Samuel, 2024) is a hybrid causal/masked language model that combines GPT-style autoregressive and BERT-style masked objectives in a single transformer stack, which is state-of-the-art on BLiMP at this scale. Baseline models are trained on the original corpus, and SAMBAL models are trained identically, but on the corpus that has been fully relexicalized according to Algorithm 1. Default SAMBAL settings are reported in Appendix A and ablations of various pipeline components are described in §5 and Appendix C.

We follow prior small-data baselines by reporting both a **short** training regime (BabyLM-style; 10 epochs over the 10M-token corpus) and a **long** training regime. The long training regime of Charpentier & Samuel (2024) runs for 7812 steps with an average batch size of around 2M tokens

(amounting to an estimated ~ 1000 epochs). Due to computational constraints, our long training regime runs for 5000 steps with an average batch size of 1.1M tokens (amounting to ~ 350 epochs and requiring 96 GPU hours). We find that, even for such small models and training data, extended large-batch training boosts performance, which accounts for the difference in our reported GPT-BERT baseline performance on BLiMP (79.2%) and that of Charpentier & Samuel (2024) (81.2%).

For all short-regime training we report mean $\pm$ standard deviation over 3 random seeds. Since syntactic benchmark performance is known to be sensitive to logit temperature, we follow Charpentier & Samuel (2024) in reporting performance at best temperature for each model.

4.2. Syntactic benchmarks

BLiMP evaluates syntax via minimal pairs, scoring whether a model prefers the grammatical sentence over a minimally different ungrammatical one. Recently, BLiMP performance of small models like GPT-BERT (81.2% with 10M tokens, 86.1% with 100M tokens) has come in range of much larger models like GPT-2 (83.0%) trained on 20B–30B tokens, while still falling short of human performance (88.6%) (Charpentier & Samuel, 2024; Warstadt et al., 2020).

SyntaxGym test suites are scored by surprisal over specified sentence regions and by whether the model satisfies a set of 2×2 inequality constraints over pairs of minimal pairs, making them less susceptible to global plausibility cues and generally more difficult (GPT-2: $\sim 78\%$). A random baseline scores $\sim 25\%$ depending on the combination of suites used. We use 25 of the 31 suites, excluding the six Garden-Path Effects suites which test human-like ambiguity resolution rather than grammaticality (Hu et al., 2020).

Table 2 reports BLiMP and SyntaxGym performance for baseline vs. SAMBAL under short and long training, with breakdowns of performance by suite in Appendix B. On BLiMP, SAMBAL matches the baseline in both regimes, while aggregate SyntaxGym performance shows a gap (66.0 vs. 70.47 in the long training regime). This discrepancy is concentrated in reflexive suites; removing these yields parity (79.8 for SAMBAL vs. 78.6 for baseline). We investigate the cause in Appendix D and summarize here:

SAMBAL behavior on SyntaxGym reflexives. SAMBAL reflexive scores collapse to near zero, and even the baseline shows a large masculine–feminine gap (e.g., `reflexive_prep`: 57.9% masc vs. 5.3% fem; `reflexive_src`: 68.4% masc vs. 26.3% fem). The collapse is driven by one recurring singular-condition pattern: in items like *The author that liked the senators hurt __*, where the antecedent (*author*) is singular, SAMBAL often prefers *themselves* over *himself/herself*.

Table 2. Syntactic benchmark performance (test accuracy %). SyntaxGym^{¬refl} removes reflexive suites.

Model	BLiMP	SG	SG ^{¬refl}
baseline (long)	79.2	70.47	78.6
SAMBAL (long)	79.3	66.0	79.8
baseline (short)	71.2 $\pm$ 0.7	54.7 $\pm$ 2.0	63.7 $\pm$ 7.8
SAMBAL (short)	71.2 $\pm$ 0.8	46.6 $\pm$ 4.4	55.9 $\pm$ 4.9

Targeted BLiMP-style diagnostics indicate that this is not a general inability to use reflexives: SAMBAL is near-ceiling when there is no distractor noun (e.g., *The manager congratulated herself/themselves*). Instead, the degradation is selective and emerges when a singular controller must be tracked across an intervening plural noun. The specific singular-controller + plural-distractor + reflexive configuration (and its reverse) that these suites test is essentially absent from our small corpus, and reflexives are skewed 2:1 towards the masculine *himself*. We interpret the baseline’s large masc–fem asymmetry as consistent with a near-threshold learning regime: there appears to be just enough evidence to partially learn the stronger generalization for masculine cases, but not enough to do so reliably for feminine cases. Small targeted augmentation largely repairs reflexive accuracy (Appendix D), indicating that the failure reflects missing coverage rather than a structural limitation.

4.3. Semantic suppression

We use targeted probes to test whether SAMBAL suppresses both aspects of semantics distinguished in §3.1: the EWoK benchmark measures *world knowledge*, and a custom swap probe targets *semantic plausibility*.

World-knowledge probe (EWoK). EWoK (Elements of World Knowledge) probes basic world knowledge using minimal pairs from 11 domains. For example: “The piano is in front of Ali. Ali turns {left/right}...The piano is {right/left} of Ali.” Even large LMs (1B–70B parameters) were found to substantially trail human performance (Ivanova et al., 2025). The baseline model attains only 54.3%, while SAMBAL scores at chance level (50.8%). Despite the small gap in overall performance, however, per-suite accuracies (Figure 2) reveal that the baseline scores above the 95% confidence level on 3 suites and in aggregate, while SAMBAL does on zero suites.

Semantic-plausibility swap probe. To isolate semantic plausibility from grammatical well-formedness in an easier, tightly controlled setting, we construct a 100-sentence swap probe where each item is defined by four content slots x_1, y_1, x_2, y_2 instantiated into a fixed syntactic template. Swapping y_1 and y_2 (or x_1 and x_2) keeps words

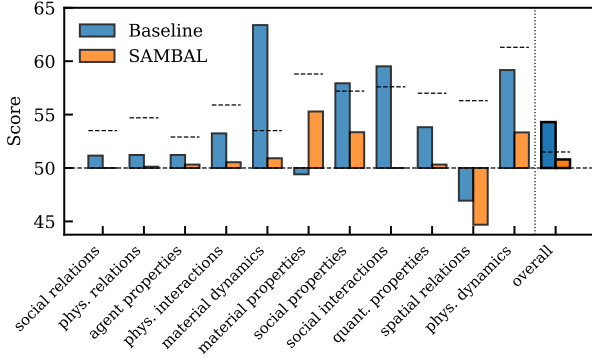


Figure 2. **World-knowledge probe (EWoK).** Accuracy by suite for SAMBAL vs. baseline. The dashed line indicates chance (50%); dotted reference lines show a 95% confidence threshold under a binomial test.

in the same syntactic roles and therefore preserves grammaticality, but is designed to disrupt plausibility, yielding a grammatical-implausible configuration (GI). Swapping $x_1 \leftrightarrow y_1$ and/or $x_2 \leftrightarrow y_2$ exchanges words across roles and yields ungrammatical-plausible (UP) and ungrammatical-implausible (UI) configurations. This construction helps to control for model sensitivity to word frequency and position (see Appendix F). For example:

template: *The boy x_1 the y_1 and x_2 the y_2 .*
vars: x_1 : sails; y_1 : boat; x_2 : catches; y_2 : mouse
GP: *The boy sails the boat and catches the mouse.*
GI: *The boy sails the mouse and catches the boat.*
UP: *The boy boat the sails and mouse the catches.*

Figure 1 plots the syntax logit margin $\Delta G = \log P(\text{grammatical}) - \log P(\text{ungrammatical})$ vs. the semantics logit margin $\Delta K = \log P(\text{plausible}) - \log P(\text{implausible})$. SAMBAL collapses ΔK toward zero, but strongly prefers the grammatical sentences, while the baseline shows large plausibility margins and no preference for grammaticality. Table 3 summarizes the same phenomenon as accuracy and mean margins, where accuracy is the fraction of templates with positive Δ .

Natural language inference (SNLI). A standard benchmark of semantic capability, SNLI (Bowman et al., 2015) tests the ability to label a premise–hypothesis pair as entailment, contradiction, or neutral. However, the task conflates the compositional and structural semantics that SAMBAL preserves with the lexical and world knowledge it removes, and carries well-documented annotation artifacts and shallow shortcuts (Gururangan et al., 2018). Some pairs are resolvable largely from structure—e.g., dropping a conjunct,

Table 3. **Swap probe: syntax vs. semantic-plausibility preferences.** Accuracy and mean margins for syntax (ΔG) and semantic plausibility (ΔK).

Model	syntax		semantic plausibility	
	Acc. (%)	ΔG	Acc. (%)	ΔK
baseline (Long)	55.0	0.65	100.0	7.62
SAMBAL (Long)	97.0	5.73	55.0	0.05

“A woman is looking into a mirror, brushing her hair” entails “A woman is brushing her hair”—whereas others require lexical or world knowledge—e.g., “A soccer game with multiple males playing” entails “Some men are playing a sport” only if one knows that soccer is a sport. Therefore, we do not treat it as a primary evaluation of semantics, and report full results in Appendix E.

4.4. Syntax–semantics disentanglement

Minimal-pair conflict benchmark construction. We test a regime where *grammaticality conflicts with an obviously plausible event interpretation*. We build a targeted grammar–plausibility *conflict set* (N=186) of minimal pairs in which the two sentences contain the same content words but differ only by swapping two noun phrases, so unigram frequency and lexical content are held constant. By construction, one member is grammatical but describes an implausible role assignment (“The cake eats the children”), while the other is plausible but contains a local agreement error (“The children eats the cake”).

To avoid confounding “entanglement” with intrinsically difficult constructions, we restrict to three local phenomena that both models handle well on BLiMP: (i) regular subject–verb agreement, (ii) determiner–noun number agreement, and (iii) passive auxiliary agreement (was/were). Each conflict item is accompanied by a matched *control* minimal pair that removes the conflict while keeping the same local grammatical dependency.

We retain items where (i) both models solve the control pair with a small positive margin, and (ii) the baseline exhibits at least a mild plausibility preference on the conflict pair. Importantly, SAMBAL’s behavior on the *conflict* comparison is *not* used for selection; it is only used to ensure that the control dependency is solvable. Because this procedure conditions on the presence of a plausibility lure for the baseline, the conflict set is baseline-conditioned by construction and should be read as an adversarial stress test of grammar–plausibility competition, not a general-purpose syntax benchmark. Our broader conclusions do not rest on it alone: the swap probe of Figure 1 and the SNLI results of §4.3 are not baseline-conditioned and point in the same direction. Further details on the benchmark are given in Appendix F.

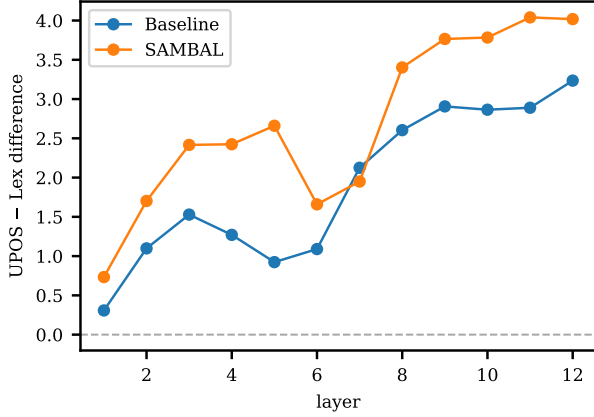


Figure 3. **Per-layer InfoNCE alignment (UPOS – Lex).** SAMBAL shows a larger syntactic-over-lexical margin across layers, consistent with reduced lexical memorization pressure.

Results. For evaluation, we combine the control and conflict sets and report the accuracy at best temperature. Both models maintain 100% accuracy on the control set, and the baseline (by design) attains only 10.2% accuracy on the conflict set, selecting the plausible member over the grammatical one on 89.8% of pairs. In contrast, SAMBAL achieves a striking 100% accuracy on the conflict set.

Representation analysis with InfoNCE. Behavioral probes suggest SAMBAL reduces plausibility reliance; we test whether this also appears in internal representations. Specifically, we ask whether hidden states become *relatively more aligned* with syntactic features (e.g., UPOS) than with lexical identity.

We compute token representations on the gold dependency parses from the Universal Dependencies English Web Treebank corpus (Nivre et al., 2020). We quantify representational alignment using an InfoNCE-style contrastive score (van den Oord et al., 2018; Poole et al., 2019) in a supervised-contrastive setup where positives share a label (Gunel et al., 2021). InfoNCE-like metrics have been used to characterize layerwise representation structure in LMs (Skean et al., 2024). We compute alignment scores for predicting (a) UPOS tags and (b) lexical identity.

Figure 3 plots the per-layer difference (UPOS alignment – Lex alignment); larger values indicate stronger emphasis on syntax relative to lexical identity. We find that this syntax-over-lexical signal increases with depth in both models, consistent with higher layers encoding more abstract structure. However, SAMBAL shows a persistent upward shift: across layers, the UPOS–Lex gap is consistently larger by roughly 25% relative to the baseline, indicating that its hidden states are systematically more aligned with syntactic categories and less dominated by lexical identity.

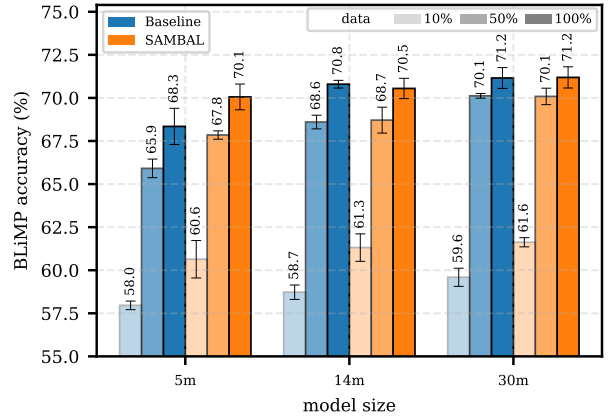
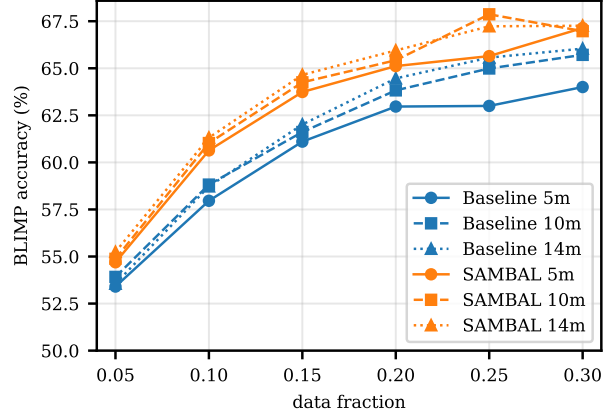


Figure 4. **Parameter and token efficiency on BLiMP.** **Top:** low-budget scaling curves up to 30% of the corpus for 5M/10M/14M models. **Bottom:** BLiMP vs. model size and data fraction (10%, 50%, 100%) for 5M/14M/30M models.

4.5. Data & parameter scaling

Figures 4–5 summarize an efficiency sweep over data budget and model size. In the low-budget regime (up to 30% of the corpus; Fig. 4 top), SAMBAL consistently shifts BLiMP scaling curves upward for 5M/10M/14M models, indicating improved per-token syntax learning. For example, for a 5M-parameter model, BLiMP increases 58.0→60.6 at 10% data and 64.0→67.1 at 30% data (corresponding means and standard deviations over 3 seeds are given in Appendix Fig. 5). Moreover, Fig. 4 (bottom) shows that these data-efficiency gains are concentrated in the smallest model and can persist even at higher budgets: for 5M, SAMBAL improves from 65.9→67.8 at 50% data and from 68.3→70.1 at 100% data, whereas once the model/data are large enough (14M/30M at 50–100%) the gap largely vanishes.

These gains also translate into *per-parameter* efficiency (Fig. 4 bottom): at the same 10% budget, a 5M SAMBAL model slightly exceeds a 30M baseline (60.6 vs. 59.6). Again, the advantage largely disappears at higher budgets

Table 4. Specialization on LoTR with LoRA fine-tuning.

Model	LoTR PPL	BLiMP Acc.
baseline (Long)	64.6	79.2
baseline (Long) fine-tuned	29.2	72.4
SAMBAL (Long)	190.4	79.3
SAMBAL (Long) fine-tuned	36.3	71.4
Best LoTR from-scratch model	55.5	53.9

and larger models, consistent with SAMBAL helping most when data and/or capacity are limiting.

Relative resource scarcity vs. absolute scale. While our experiments are small scale, we have some indication that efficiency benefits may be more linked to relative scarcity rather than absolute scale. Relating SAMBAL’s BLiMP advantage to a *data-utilization* measure (the baseline’s accuracy as a fraction of the best SAMBAL accuracy at the same budget), the advantage is strongly predicted by how much headroom remains ($R^2 = 0.76$), and points from different data budgets intermingle rather than stratifying by absolute token count: the gain appears whenever the model is parameter-constrained *relative* to its data budget (Appendix G).

4.6. Small-domain adaptation

Can a SAMBAL model rapidly acquire domain content after pretraining? The use case is modular: SAMBAL pretraining encodes syntax in the base weights, and a lightweight LoRA adapter adds domain knowledge (only the adapter is trained)—targeting cases where one wants a linguistically-capable model that knows little beyond a target domain too small to train from scratch. We perform LoRA fine-tuning on the *Lord of the Rings* trilogy (580k words) (Tolkien, 1994). As expected, SAMBAL begins with much worse domain perplexity than baseline (190.4 vs. 64.6) but adapts quickly with limited exposure (36.3 vs. 29.2 after fine-tuning). The best model trained from-scratch on LoTR obtains PPL 55.5, indicating that specialization benefits from a syntax-competent initialization. Table 4 compares LoTR perplexity and BLiMP accuracy before vs. after specialization; we see that both SAMBAL and baseline models lose around 7 points on BLiMP after fine-tuning. For comparison, the model trained on LoTR alone achieves only 53.9% on BLiMP.

Similar results hold for a more technical domain of the same size—a collection of PubMed abstracts (NLM, 2024); see Appendix J.

LoTR generations. Qualitatively, generations reflect the same pattern. Before specialization, SAMBAL tends to produce grammatical but semantically unusual continuations; after LoRA fine-tuning it begins to use domain-appropriate

entities, themes and stylistic cues:

‘It’s a dangerous business, Frodo, going out of your door’ he used to say.

SAMBAL: *Marlon went out comfortably before seeing, “How does that find our media?” To which the husband survived curiously and said, “Well, well...the one thing I meet is how we look and find with each other. When someone reaches personal surgery at a potato year..*

SAMBAL ft: *‘We shall be too good here now.’ The hobbits looked at him sadly and silently. Sam sat up, his face grappled with feeling that the hobbit was forgetting the sound of eating and singing softly.*

We include generations of the baseline model and additional SAMBAL generations in Appendix I.

5. Discussion

Limitations, leakage, and ablations. Even with constrained relexicalization, some semantic traces may remain in SAMBAL text (e.g., function-word distributions, constructions correlated with meaning, or distributional regularities induced by replacement statistics). We also preserve certain boundary features such as gender/humanness/animacy when they directly affect acceptability (agreement and binding), even though they correlate with meaning. Ablations clarify the role of these choices: ablating gender reduces SAMBAL-short BLiMP from 71.2 to 70.47, while ablating humanness reduces it to 69.86; on reflexive/anaphor gender-agreement tasks, dropping gender collapses performance to approximately chance. Removing corpus-conditioned sampling drops BLiMP to 66.42, so context-conditioned candidate selection is important for grammatical fit and coverage.

All of these components—residual function-word and constructional patterns, the retention of meaning-adjacent boundary features (gender/humanness), and corpus-conditioned candidate selection—are plausible pathways by which distributional world-knowledge regularities could re-enter despite constrained relexicalization. SAMBAL is thus more precisely described as preserving syntax plus a thin layer of meaning-adjacent information than as a perfectly clean syntax/semantics separation; we view the delineation of this boundary—what to preserve, what to ablate, and why (Table 1, Appendix K)—as itself a contribution. Our no-knowledge probes indicate that plausibility and encyclopedic cues are substantially suppressed, but they do not rule out subtler leakage channels. A natural direction for future work is adversarial relexicalization: constructing transformations that intentionally reintroduce spurious form–semantics correlations between form and latent semantics/domain while preserving grammaticality, to better

characterize when SAMBAL succeeds or fails at suppressing non-syntactic cues.

Another limiting factor is SAMBAL’s reliance on high-resource English tooling (parsing, morphology, external lexicons), and these resources can be incomplete or idiosyncratic: permissive matching admits odd outputs, while strict matching can yield empty candidate pools. Some constructions we wish to control lack reliable off-the-shelf inventories (e.g., NPI licensors or fine-grained *to*-infinitival verb/adjective classes), motivating custom lexica and bucket definitions; portability to lower-resource languages remains open.

Why might SAMBAL improve efficiency? A working hypothesis is that SAMBAL improves data/compute efficiency by suppressing selectional preferences and plausibility priors, shifting training pressure toward morphosyntax, licensing, and structural dependencies. Future work could test this by mapping which phenomena benefit most and by varying the availability of semantic signals while holding grammatical constraints fixed.

Another possible source of the efficiency gain is the replacement lexical distribution: SAMBAL draws replacements from a reduced vocabulary, which could yield a “simpler” lexical distribution than the baseline’s. Estimates of corpus entropy show this is only partially true: unigram entropy H_1 is indeed slightly lower (10.45 vs. the baseline’s 10.80 bits/word), but conditional bigram entropy h_2 is actually *higher* (+0.57 bits/word), as expected when co-occurrence structure is ablated. An additional control experiment also points away from a pure lexical-entropy explanation: training on original text filtered to SAMBAL’s top-25k vocabulary—an intervention that reduces H_1 further than SAMBAL ($H_1 = 10.16$) while leaving h_2 essentially unchanged, and preserves meaning—lands above the baseline but below SAMBAL at the 10% token budget. This indicates that simpler lexical distribution could explain part but not all of SAMBAL’s advantage (Appendix H).

Pushing this simplification further exposes a tradeoff between syntactic isolation and lexical coverage. Collapsing each replacement class to a single representative would maximize structure-only training but cripple real-vocabulary evaluation and downstream adaptation; even the milder step of restricting replacements to the small BLiMP vocabulary improves SAMBAL-short BLiMP by about 2 points while lowering SyntaxGym by about 2 and raising perplexity after LoTR adaptation. Natural-language evaluation and adaptation metrics are therefore useful controls against gains on form-focused benchmarks that merely reflect drift toward an overly simplified lexical distribution. Systematically exploring the choice of replacement vocabulary—tracing out the grammar/coverage/adaptation tradeoffs as it is tightened

or broadened—is a promising direction for future work.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Implementation details for SAMBAL

SAMBAL transforms natural sentences into *grammatical nonsense* via constrained relexicalization: we replace eligible *open-class* tokens (content nouns/verbs/adjectives/adverbs and proper nouns) with new lemmas drawn from syntactically compatible candidate pools, then inflect the sampled lemma to match the original token’s morphosyntactic realization. The design goal is to preserve grammatical structure (agreement, morphosyntax, and coarse complement structure / c-selection) while ablating semantic and world-knowledge content.

Primary configuration used in our experiments. Unless explicitly stated otherwise, we run SAMBAL with the following default settings:

- **Parser / tagger:** a transformer-based English pipeline (spaCy `en_core_web_trf` (Honnibal et al., 2020)) provides tokenization, POS tags (PTB), morphology, and dependencies.
- **Replacement scope:** proper nouns are replaced; pronouns are preserved. Closed-class tokens are preserved (see below).
- **Humanness and gender preservation:** Human-denoting heads are replaced with other human-denoting heads, and gender-marked words/names are replaced within the same gender class.
- **Vocabulary gating:** swap candidates with a minimum frequency threshold of 1 in the training corpus are allowed.
- **Context-conditioned sampling:** uses frequency-weighted sampling from corpus-derived context buckets, with (i) a per-bucket minimum lemma count of 2 and a backoff policy requiring at least 2 surviving candidates and at least 5% of the unconditioned pool before accepting a bucket.
- **To-infinitive safety:** In raising/control/ECM-sensitive infinitival environments, we restrict replacements using dedicated lexicons; if no safe compatible pool is available, we freeze the governing head rather than falling back to generic sampling.
- **Round-trip validation:** We reparse the augmented output and reject–resample 4 times if structural invariants fail (details below).
- **VerbNet soft-preposition fallback:** enabled with a minimum pool size threshold of 10. For plain intransitive verbs with adjunct PPs, if enforcing the observed governed preposition would shrink the VerbNet candidate set below 10, we ignore the preposition in VerbNet matching (while still retaining it as a contextual feature for corpus-based gating).

External resources and derived inventories. SAMBAL relies on a set of static inventories and lexicon:

- **WordNet:** provides large lemma inventories for nouns, adjectives, and adverbs (used as base replacement pools) and supports additional coarse syntactic checks for verbs (e.g., minimum object requirements and complement-type compatibility) (Miller, 1995)
- **VerbNet:** provides verb classes and frame information used to bucket verbs by coarse subcategorization signatures.
- **Countability lexicon for nouns derived from Wiktionary:** supplies coarse noun classes *mass*, *count*, *both*, and *plural-only* to preserve determiner/number compatibility (see below). (Wiktionary)
- **Function-word list (UD English Web Treebank):** extracted from UD-EWT tokens whose universal POS is in {ADP, DET, AUX, PRON, PART, SCONJ, CCONJ}, providing a high-coverage function-word inventory beyond stopword heuristics. (Universal Dependencies, 2023)
- **Multiword-expression (MWE) inventories:** (i) fixed MWEs harvested from UD-EWT `fixed` relations, (ii) verb–particle constructions from UD-EWT (`compound:prt`), and (iii) verbal MWE patterns derived from STREUSLE (compiled into matcher patterns). These spans are preserved as atomic units during augmentation. (Universal Dependencies, 2023; Schneider & Smith, 2018)
- **Negative polarity items (NPIs):** a Wiktionary-derived list of English NPIs is used both to (i) preserve NPI tokens themselves and (ii) prevent NPIs from being introduced as replacement candidates.

- **NPI licensor patterns:** a hand-curated inventory of token- and dependency-based licensor patterns marks additional protected spans (e.g., negation-like and quantificational licensors).
- **Human noun inventory:** derived from WordNet `noun.person` (plus associated multiword human expressions), enabling human→human replacement.
- **Names and proper nouns:** U.S. SSA baby names are used to construct given-name pools partitioned into male/female/neutral (based on aggregate frequency and dominance criteria). We additionally maintain a training-corpus-frequency derived inventory of allowable single-token proper nouns for replacement.
- **Gendered common words:** a filtered version of the `ecmonsens/gendered_words` lexicon supplies unambiguous gender labels for common nouns; we prefilter it against the allowed vocabulary. (ecmonsens, 2023)
- **Allowed vocabulary:** an inventory of surface tokens extracted from the target pretraining corpus using the same tokenizer/parser, used to gate substitutions so that realized outputs stay in-vocabulary.
- **Context statistics:** corpus-derived counts of lemma occurrences conditioned on syntactic context buckets, used for context-conditioned gating and frequency-weighted sampling. We build these statistics by parsing the corpus and tabulating lemma counts keyed by compact context signatures; to control size we restrict collection to a fixed high-frequency word list, the top-25k words according to wordfreq python library (wordfreq developers, 2022).

Protected tokens and spans. Before sampling any replacements, we mark protected items and leave them unchanged. Protection includes: (i) closed-class tokens (function words, auxiliaries, complementizers, determiners, adpositions, conjunctions, etc.), (ii) any token in the function-word inventory or stopword set, (iii) NPI tokens and detected licensor spans (both token-based and dependency-pattern matches), (iv) any token inside a detected MWE span (fixed MWEs and verbal MWEs), and (v) structural backbones such as lexical *be* and auxiliary/copular heads in sensitive configurations. We also conservatively protect tokens that behave as grammar-critical operators in minimal-pair settings (e.g., quantificational/degree constructions), including a WordNet-derived set of quantity nouns used to freeze binominal degree MWEs (e.g., *a lot of*, *a number of*).

Replacement pools and constraints by category. Replacements are defined at the *lemma* level; we then realize a surface form that matches the original token’s PTB tag and morphosyntactic profile.

Nouns (common nouns). We bucket noun lemmas by coarse countability class. At runtime, we select a noun pool compatible with the slot’s number and determiner context:

- *Bare singular* heads (singular common nouns without an overt determiner) are only replaced with lemmas that can appear bare (typically mass-capable or otherwise licensed).
- *Plural* heads are replaced with lemmas compatible with plural realization, including plural-only nouns.
- *Determiner-sensitive* contexts respect mass/count behavior so that replacements do not force impossible determiner patterns.

When humanness preservation is enabled, human-denoting noun heads are replaced from a human-only pool; multiword human expressions are preserved as spans.

Proper nouns. We replace each eligible proper-noun token with a sampled name while preserving casing. If the token matches a known given name with a gender label, we sample a replacement from the same gender pool; otherwise we sample from the allowed-proper-noun inventory.

Verbs. We compute a coarse *subcategorization signature* from the dependency parse and retrieve verb candidates from VerbNet buckets keyed by:

- *Objecthood:* intransitive vs. transitive vs. ditransitive structure (with passives handled explicitly).
- *Complement kind:* none vs. NP object vs. double-object vs. clausal complements (that/wh), infinitival complements (to/for-to), and gerundive complements.

- *Governed prepositions*: observed governed prepositions are treated as selectional constraints when they plausibly reflect argument structure; for adjunct-like PP attachments under plain intransitives we apply the soft-preposition fallback described above.

We then apply conservative additional guards to avoid known failure modes: we freeze auxiliaries/copulas and *be*, avoid swapping heads in expletive/ECM-like configurations, and freeze phrasal-verb heads (particle compatibility is not guaranteed). We further filter candidates using WordNet-derived syntactic-frame checks for (i) minimum object requirements in transitive slots, (ii) compatibility with clause and infinitival complement types, and (iii) compatibility with specific PP head types when required.

To-infinitive control/raising/ECM environments (optional but enabled by default). Some heads differ systematically in whether and how they license infinitival complements and argument realization. When a verb or adjective participates in a raising/control/ECM-sensitive infinitival spine, we restrict candidates using dedicated lexicons that encode a small set of *license tags* (e.g., subject-control vs. object-control vs. raising vs. ECM, plus expletive *it/there* and passive variants). Only lemmas whose licenses subsume the local surface pattern are eligible. If a compatible pool cannot be formed, we freeze the governing head (rather than falling back to generic VerbNet/WordNet sampling).

Adjectives. Adjective candidates are drawn from a WordNet-derived adjective inventory, filtered to avoid proper/demonym-like items. In infinitival-adjective spines (e.g., *easy to V*), we again apply license-based filtering using the to-infinitive adjective lexicon, including special handling for stranded-preposition (“gap-PP”) patterns.

Adverbs. Adverb candidates come from a WordNet-derived adverb inventory that is prefiltered to items that tag as plain adverbs (RB) under the parser’s tagger (to avoid degree-morphology edge cases by default).

Context-conditioned candidate gating and sampling. To reduce parser-fragility and improve surface plausibility without reintroducing semantic content, we condition replacements on corpus-derived syntactic context statistics. For each replaceable token, we compute a compact *context bucket key* composed of: (i) POS and PTB tag, (ii) dependency role and head POS/tag, (iii) local neighborhood cues (nearby POS tags and a coarse relative-position feature), (iv) a compact morphological signature, and (v) verb-specific frame features (coarse valency, complement kind, and governed-preposition set). We intersect the lexicon-derived candidate set with the set of lemmas observed in that context bucket in the base corpus and sample replacements proportionally to their observed bucket frequencies. If the intersection is too small (absolute and relative thresholds given in the primary configuration above), we back off through a deterministic chain of progressively coarser buckets (dropping some contextual features) until we obtain a sufficiently large candidate set.

Surface realization and post-processing. After sampling a replacement lemma, we inflect it to match the original token’s PTB tag using a morphology engine (lemmatization/inflection for nouns and verbs, with conservative fallbacks). We preserve orthographic casing by matching the original token’s capitalization pattern. Finally, we apply a local determiners fix for *a/an* to ensure that the indefinite article matches the pronunciation of the following word after substitution.

Round-trip validation (UD reparsing). When enabled, we reparse the augmented output and require it to preserve a small set of robust, grammar-relevant invariants. Concretely, we compare: (i) the multiset of coarse verb-frame keys (intrans/trans/ditrans), (ii) the counts of expletive constructions (*there* vs. *it*), (iii) the counts of tough-like vs. raising-like infinitival adjective patterns under copular *be*, and (iv) a conservative count of ECM/expletive-raising configurations. If any invariant differs, we reject the sample and resample replacements (up to the configured maximum number of retries), accepting the first augmentation that passes the checks.

Parser quality and manual validation. Because relexicalization is conditioned on the initial parse, tagging and dependency errors can propagate downstream. The `en_core_web_trf` pipeline reports high accuracy (roughly 0.98 POS, 0.95 UAS, 0.94 LAS) (Honnibal et al., 2020), and our pipeline introduces occasional realization errors of its own. To gauge the net fidelity of the transformed corpus, we manually spot-checked 100 sampled SAMBAL sentences and found 87% to be fully grammatical. Despite this imperfection, we take the downstream BLiMP and SyntaxGym results as evidence that this level of fidelity is sufficient for our purposes.

B. Full BLiMP and SyntaxGym suite results

Table 5. BLiMP per-suite accuracy (%).

BLiMP paradigm	baseline	SAMBAL
ANAPHOR AGREEMENT		
anaphor_gender_agreement	89.7	94.1
anaphor_number_agreement	94.9	96.6
ARGUMENT STRUCTURE		
animate_subject_passive	78.6	77.2
animate_subject_trans	87.3	88.8
causative	70.1	50.7
drop_argument	67.5	71.4
inchoative	68.4	58.1
intransitive	75.4	72.6
passive_1	88.5	81.4
passive_2	87.6	86.7
transitive	82.4	64.8
BINDING		
principle_A_c_command	46.7	57.6
principle_A_case_1	99.9	99.7
principle_A_case_2	96.2	88.0
principle_A_domain_1	84.4	91.1
principle_A_domain_2	59.8	58.9
principle_A_domain_3	55.9	59.4
principle_A_reconstruction	26.4	37.9
CONTROL/RAISING		
existential_there_object_raising	81.3	63.7
existential_there_subject_raising	86.5	78.0
expletive_it_object_raising	66.9	59.6
tough_vs_raising_1	74.4	66.2
tough_vs_raising_2	78.0	81.4
DETERMINER-NOUN AGR.		
determiner_noun_agreement_1	98.7	97.0
determiner_noun_agreement_2	99.1	98.0
determiner_noun_agreement_irregular_1	89.1	89.7
determiner_noun_agreement_irregular_2	96.4	93.7
determiner_noun_agreement_with_adjective_1	96.5	94.6
determiner_noun_agreement_with_adj_2	97.2	95.1
determiner_noun_agreement_with_adj_irregular_1	84.5	90.7
determiner_noun_agreement_with_adj_irregular_2	95.3	91.7
ELLIPSIS		
ellipsis_n_bar_1	76.6	76.9
ellipsis_n_bar_2	96.3	74.1
FILLER-GAP		
wh_questions_object_gap	70.9	74.4
wh_questions_subject_gap	83.5	81.5
wh_questions_subject_gap_long_distance	78.7	73.0
wh_vs_that_no_gap	90.2	87.2

Continued on next page

BLiMP paradigm	baseline	SAMBAL
wh_vs_that_no_gap_long_distance	94.1	93.8
wh_vs_that_with_gap	89.2	93.2
wh_vs_that_with_gap_long_distance	59.9	61.4
IRREGULAR FORMS		
irregular_past_participle_adjectives	96.6	99.4
irregular_past_participle_verbs	86.5	85.7
ISLAND EFFECTS		
adjunct_island	80.8	91.3
complex_NP_island	41.0	54.4
coordinate_structure_constraint_complex_left_branch	76.2	94.6
coordinate_structure_constraint_object_extraction	92.9	93.1
left_branch_island_echo_question	21.9	20.2
left_branch_island_simple_question	90.5	97.5
sentential_subject_island	35.0	50.1
wh_island	75.0	90.1
NPI LICENSING		
matrix_question_npi_licensor_present	80.0	84.1
npi_present_1	51.1	86.4
npi_present_2	57.8	65.3
only_npi_licensor_present	95.3	99.0
only_npi_scope	85.5	82.6
sentential_negation_npi_licensor_present	98.3	94.8
sentential_negation_npi_scope	78.5	59.4
QUANTIFIERS		
existential_there_quantifiers_1	98.0	98.1
existential_there_quantifiers_2	26.2	14.3
superlative_quantifiers_1	57.7	100.0
superlative_quantifiers_2	79.4	78.4
SUBJECT-VERB AGR.		
distractor_agreement_relational_noun	91.2	88.5
distractor_agreement_relative_clause	82.6	85.3
irregular_plural_subject_verb_agreement_1	94.4	87.2
irregular_plural_subject_verb_agreement_2	90.9	87.1
regular_plural_subject_verb_agreement_1	90.0	84.9
regular_plural_subject_verb_agreement_2	90.1	93.8

Table 6. SyntaxGym per-suite accuracy (%)

SyntaxGym suite	baseline	SAMBAL
number_orc	47.37	68.42
number_prep	63.16	84.21
number_src	68.42	57.89
reflexive_orc_fem	15.79	0.00
reflexive_orc_masc	31.58	0.00
reflexive_prep_fem	5.26	0.00
reflexive_prep_masc	57.89	5.26
reflexive_src_fem	26.32	0.00
reflexive_src_masc	68.42	0.00
npi_orc_any	86.84	89.47
npi_orc_ever	94.74	94.74
npi_src_any	65.79	84.21
npi_src_ever	100.00	97.37
fgd_hierarchy	20.83	10.42
fgd_object	66.67	79.17
fgd_pp	37.50	54.17
fgd_subject	37.50	45.83
center_embed	85.71	64.29
center_embed_mod	75.00	57.14
cleft	97.50	100.00
cleft_modifier	97.50	95.00
subordination	100.00	95.65
subordination_orc-orc	100.00	100.00
subordination_pp-pp	100.00	95.65
subordination_src-src	95.65	95.65

C. Ablation table

Table 7. BLiMP ablation results.

Ablation	BLiMP
SAMBAL (Short)	71.2 ± 0.8
no to-inf	70.65 ± 0.30
no gender	70.47 ± 0.24
no UD-roundtrip	70.38 ± 0.42
no humanness	69.86 ± 0.86
no context buckets	66.42 ± 0.78

D. SAMBAL reflexive performance exploration and diagnosis

What goes wrong. SyntaxGym reflexive suites are an outlier in our evaluation: SAMBAL collapses to near-zero suite accuracy, and even the baseline shows a large masculine–feminine gap (e.g., `reflexive_prep`: 57.9% masc vs. 5.3% fem; `reflexive_src`: 68.4% masc vs. 26.3% fem). Inspecting condition-level behavior, we found that SAMBAL’s collapse is driven by a single recurring error in the *singular* condition: in items like *The author that liked the senators hurt __*, where the true antecedent (*author*) is singular but a plural noun intervenes (*senators*), SAMBAL often prefers *themselves* over *himself/herself*. Because SyntaxGym suites require all expected comparisons to hold simultaneously, consistently failing this singular contrast is sufficient to drive near-zero suite scores.

Targeted BLiMP-style diagnostics. To localize the failure, we constructed BLiMP-format minimal pairs that separate interference from baseline lexical preferences. The results show a sharply selective deficit: (i) SAMBAL is near-ceiling when interference is removed (100% on singular-attractor controls), (ii) SAMBAL passes fixed-form licensing controls (e.g., *himself*-only and *themselves*-only number tests are both >90%), but (iii) SAMBAL drops substantially when a singular controller must be tracked across an intervening plural noun (32.22% on the plural-attractor diagnostic at T=1.0; 63.33% at the best temperature), matching the singular-contrast failure observed in SyntaxGym.

Corpus-side evidence and why we treat this as near-threshold. We examined if this failure can be traced to surface-level statistics of the training corpus. The precise singular-controller + plural-distractor + reflexive configuration (and its reverse) that the SyntaxGym suites test is essentially absent from our corpus: of 5,069 extracted reflexive occurrences, only 39 fall in the broad neighborhood of this pattern (singular controller, intervening plural noun, reflexive), and these belong to a wider family of cue-conflict cases that do not teach the exact tested contrast. This sparsity, together with the baseline’s already-large masc–fem asymmetry on SyntaxGym reflexives, is consistent with a near-threshold learning regime in which there is limited and uneven signal for the stronger long-distance generalization (enough to partially learn the masculine case, but not reliably for the feminine case). Corpus statistics for these patterns are very similar between the original and SAMBAL-ablated corpora, so we do not attribute SAMBAL reflexive performance to a large distributional shift; however, in such a data-sparse regime, even small degradations in effective learnability (e.g., slightly harder retrieval conditions or noisier controller tracking under ablation) could plausibly push performance over a learnability edge.

Targeted augmentation repairs performance. To test whether the collapse reflects a structural inability to learn long-distance binding or merely missing pattern coverage, we augmented the training corpus with relevant examples. Specifically, we created two SAMBAL-ablated sentences from each of the 228 grammatical SyntaxGym reflexive items and (because reflexive-only augmentation slightly hurt number-suite behavior) added analogous examples from the number-suite items as well, increasing the corpus by only about 6,000 tokens. Training short-regime models on the augmented SAMBAL corpus raises the reflexive-suite average to 98.0%, with overall SyntaxGym improving to 66.8 ± 6.9 . Augmenting the baseline corpus with the same examples yields 100% on the reflexive suites and 73.8 ± 1.3 on SyntaxGym overall. We therefore interpret the collapse as a genuine weak point of the 10M-token regime that is driven by missing pattern coverage rather than by an inability of SAMBAL-trained models to learn long-distance binding in principle: the relevant pattern is adversarial and uncommon in natural corpora, the 10M-token corpus is near-threshold for it, and modest targeted augmentation largely repairs it.

E. SNLI evaluation details

We evaluate on SNLI (Bowman et al., 2015), classifying whether a hypothesis is entailed by, contradicts, or is neutral to a premise (chance = 33.3%). Table 8 reports full-test and SNLI-hard accuracy. The above-chance SAMBAL score should be read against SNLI’s well-documented shallow shortcuts: a hypothesis-only classifier reaches ~67% (Gururangan et al., 2018) and a word-overlap baseline 50.4% (Bowman et al., 2015). The artifact-controlled SNLI-hard subset (Gururangan et al., 2018) removes hypothesis-only artifacts; there the baseline obtains 44.3% and SAMBAL 37.5%—much closer to chance—while preserving the ~7pp SAMBAL–baseline gap seen on the full set, consistent with SAMBAL retaining little broad semantic competence beyond what shallow cues afford. The variant trained without context-conditioned sampling (59.7%) essentially matches standard SAMBAL (60.1%), indicating that context-conditioned sampling injects little semantic content.

Table 8. SNLI results (accuracy %). “no ctx. samp.” is the SAMBAL variant without context-conditioned sampling.

Model	SNLI	SNLI-hard
baseline	67.4	44.3
SAMBAL	60.1	37.5
SAMBAL (no ctx. samp.)	59.7	—
chance	33.3	33.3

F. Probe suite construction details

Here we detail how we construct and score the two probe sets used in §4.

- the *semantic-plausibility swap probe* (four content slots in a fixed template; syntax vs. plausibility preferences), and
- the *grammar-plausibility conflict benchmark* (role-reversal minimal pairs with local agreement errors).

F.1. Semantic-plausibility swap probe: four-slot construction

Goal. The swap probe isolates a model’s preference for (a) grammatical well-formedness vs. (b) semantic plausibility, while controlling lexical identity and (to a large extent) unigram frequency effects by reusing the same content words across derived variants.

Template and slots. Each item is defined by four open-class slots (x_1, y_1, x_2, y_2) instantiated into a fixed syntactic template with two transitive predicates. We evaluate *both* conjunct orders to control for position/order effects:

$$\begin{aligned} T^{12}(a, b, c, d) &= \text{The boy } a \text{ the } b \text{ and } c \text{ the } d. \\ T^{21}(a, b, c, d) &= \text{The boy } c \text{ the } d \text{ and } a \text{ the } b. \end{aligned}$$

Here x_1, x_2 are transitive verbs (surface-inflected to match the template), and y_1, y_2 are count nouns compatible with *the*. We construct items so that the base pairing (x_1, y_1) and (x_2, y_2) yields an intuitively plausible base sentence.

Derived variants via swaps (with both conjunct orders). From each slot quadruple we deterministically generate four configurations (GP, GI, UP, UI), and for each configuration we instantiate two sentence variants corresponding to the two conjunct orders (12 and 21). This yields 8 sentences per item.

1. **GP (grammatical-plausible).** The base instantiation:

$$\text{GP}^{12} = T^{12}(x_1, y_1, x_2, y_2), \quad \text{GP}^{21} = T^{21}(x_1, y_1, x_2, y_2).$$

2. **GI (grammatical-implausible).** Swap the two objects, keeping words in their original syntactic roles:

$$\text{GI}^{12} = T^{12}(x_1, y_2, x_2, y_1), \quad \text{GI}^{21} = T^{21}(x_1, y_2, x_2, y_1).$$

This preserves grammaticality by construction but is intended to disrupt plausibility by mismatching verb-object pairings.

3. **UP (ungrammatical-plausible).** Swap each verb with its corresponding object (crossing syntactic roles):

$$\text{UP}^{12} = T^{12}(y_1, x_1, y_2, x_2), \quad \text{UP}^{21} = T^{21}(y_1, x_1, y_2, x_2).$$

This keeps the same multiset of content words and largely preserves the intended event associations at the bag-of-words level, while introducing a clear category/word-order violation.

4. **UI (ungrammatical-implausible).** Apply both perturbations (role-crossing swap plus object swap):

$$\text{UI}^{12} = T^{12}(y_2, x_1, y_1, x_2), \quad \text{UI}^{21} = T^{21}(y_2, x_1, y_1, x_2).$$

Order-averaged scores. Let $s_m(\cdot)$ denote the sentence score for model m . For each configuration $C \in \{\text{GP}, \text{GI}, \text{UP}, \text{UI}\}$, we define an order-averaged score:

$$\bar{s}_m(C) = \frac{1}{2} \left(s_m(C^{12}) + s_m(C^{21}) \right).$$

This averaging reduces sensitivity to conjunct order and token-position effects.

Margins and accuracies (averaging over the two order variants). We define a *semantic plausibility* margin by comparing grammatical-plausible vs. grammatical-implausible, averaging over both conjunct orders:

$$\Delta_K^m = \bar{s}_m(\text{GP}) - \bar{s}_m(\text{GI}) = \frac{1}{2} \sum_{o \in \{12, 21\}} s_m(\text{GP}^o) - \frac{1}{2} \sum_{o \in \{12, 21\}} s_m(\text{GI}^o). \quad (1)$$

We define a *syntax/grammar* margin by comparing *all* grammatical sentences (GP and GI in both orders) against *all* ungrammatical sentences (UP and UI in both orders):

$$\Delta_G^m = \frac{1}{4} \sum_{o \in \{12, 21\}} \left(s_m(\text{GP}^o) + s_m(\text{GI}^o) \right) - \frac{1}{4} \sum_{o \in \{12, 21\}} \left(s_m(\text{UP}^o) + s_m(\text{UI}^o) \right) \quad (2)$$

$$= \frac{1}{2} \left(\bar{s}_m(\text{GP}) + \bar{s}_m(\text{GI}) \right) - \frac{1}{2} \left(\bar{s}_m(\text{UP}) + \bar{s}_m(\text{UI}) \right). \quad (3)$$

We report “plausibility accuracy” as $\mathbf{1}[\Delta_K^m > 0]$ averaged over items and “syntax accuracy” as $\mathbf{1}[\Delta_G^m > 0]$ averaged over items; we also report mean margins $\mathbb{E}[\Delta_K^m]$ and $\mathbb{E}[\Delta_G^m]$.

Item selection. We construct a fixed set of 100 slot-quadruples by choosing common transitive verbs and concrete nouns so that (i) the base sentence is intuitively plausible and (ii) the swapped-object variant is anomalous while remaining grammatical.

F.2. Grammar–plausibility conflict benchmark: role-reversal swaps

Goal. This benchmark targets settings where grammatical well-formedness conflicts with obvious semantically plausible interpretation. Each item is a *swap-only* minimal pair: the two sentences contain the same content words and differ only by swapping two noun phrases, which controls unigram identity while flipping thematic roles. Grammaticality is determined locally by an agreement dependency.

Template families (local, high-BLiMP phenomena). To avoid confounding “entanglement” with intrinsically difficult constructions, we restrict to short templates built from three local phenomena that both models handle well on BLiMP:

1. **SVA role reversal (regular subject–verb agreement).** Let A_{pl} be an agent-like plural noun (e.g., *children*) and B_{sg} a patient-like singular noun (e.g., *cake*), and let $V_{3\text{sg}}/V_{\text{pl}}$ be the corresponding verb forms.

GOOD (grammatical, implausible) : *The B_{sg} $V_{3\text{sg}}$ the A_{pl} .*

BAD (ungrammatical, plausible) : *The A_{pl} $V_{3\text{sg}}$ the B_{sg} .*

2. **Det–noun role reversal (determiner–noun number agreement).** We use a singular-only determiner (e.g., *each* or *a*) and a flexible determiner (e.g., *some*) with a past-tense verb to keep verb morphology neutral:

GOOD : *Each B_{sg} V_{past} some A_{pl} .*

BAD : *Each A_{pl} V_{past} some B_{sg} .*

3. **Passive role reversal (auxiliary agreement in passives).** With past participle V_{pp} :

GOOD : *The A_{pl} were V_{pp} by the B_{sg} .*

BAD : *The B_{sg} were V_{pp} by the A_{pl} .*

In all families, the GOOD and BAD sentences differ only by swapping A_{pl} and B_{sg} ; grammaticality is determined by the agreement morphology ($V_{3\text{sg}}$ vs. plural subject, singular determiner vs. plural head, or *was/were*).

Controls (to ensure the phenomenon is easy). Each conflict item is paired with a matched *control* minimal pair that removes the grammar–plausibility conflict while keeping the same local agreement dependency. Concretely, we reuse the conflict BAD sentence as `control_bad`, and form `control_good` by keeping the *plausible* role assignment but correcting only the agreement morphology (e.g., *The cake was eaten by the children* vs. *The cake were eaten by the children*). This ensures that any errors on the conflict set are not simply due to failure to represent the underlying agreement rule.

Candidate generation. We generate a large pool of candidates by sampling from a small inventory of typed micro-frames: agent-like plural subjects (animate groups), patient-like singular objects (typically inanimate), and strongly agentive verbs with the required inflections (V_{pl} , V_{3sg} , V_{past} , V_{pp}). The grammatical label is determined purely by the template.

Automatic scoring and filtering (stress-test slice). We score all sentences with $\text{temperature}=1$. Let $\Delta_{\text{ctrl}}^m = s_m(\text{control_good}) - s_m(\text{control_bad})$ be the control margin for model m , and let $\Delta_{\text{conf}}^{\text{base}} = s_{\text{base}}(\text{BAD}) - s_{\text{base}}(\text{GOOD})$ be the baseline’s preference for the plausible-but-ungrammatical member. We retain candidates that satisfy:

1. **Control solvability (both models):** $\Delta_{\text{ctrl}}^{\text{base}} \geq \tau$ and $\Delta_{\text{ctrl}}^{\text{SAMBAL}} \geq \tau$, where $\tau = 0.1$.
2. **Presence of a plausibility lure (baseline only):** $\Delta_{\text{conf}}^{\text{base}} \geq \gamma$, where $\gamma = 0.1$.

We then subsample a fixed number per family with simple de-duplication constraints (e.g., not repeating the same noun pair/verb frame excessively). Importantly, SAMBAL’s behavior on the *conflict* comparison is not used for selection; it is used only to ensure that the control dependency is solvable. Because the filtering conditions on the existence of a plausibility lure for the baseline, the resulting conflict set should be interpreted as *susceptibility under adversarial tension* rather than as a general-purpose syntax benchmark.

G. Additional scaling results

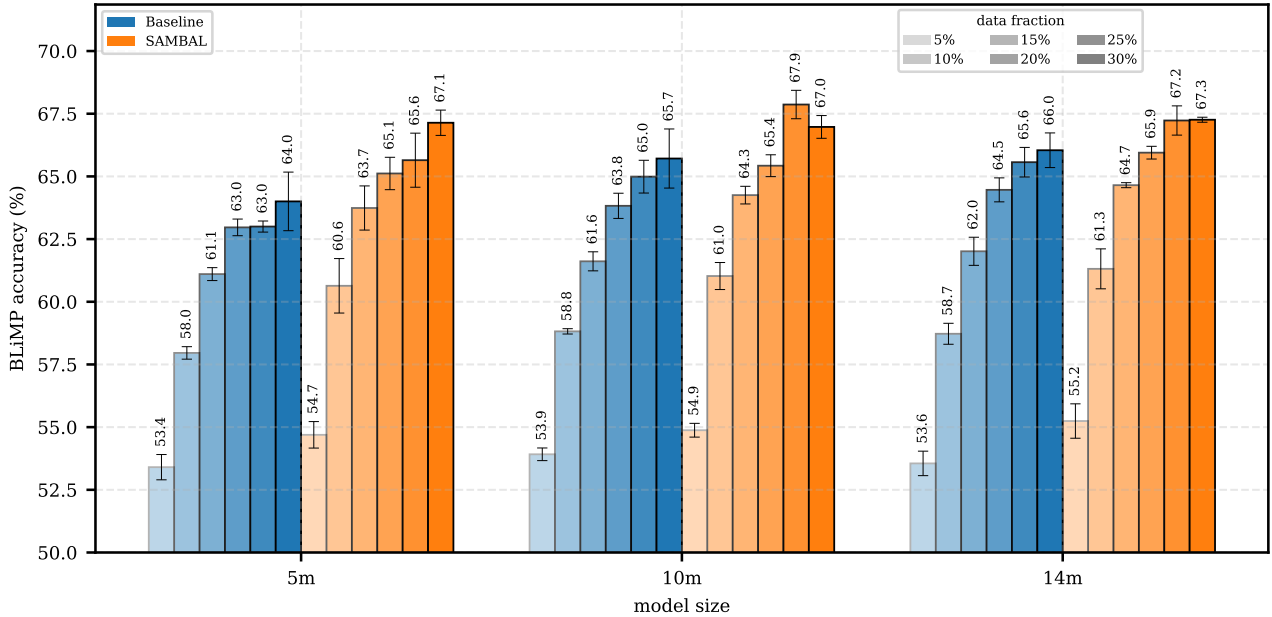


Figure 5. **Parameter and token efficiency.** BLiMP vs model size and data fraction for 0.5M–3M tokens

Relative resource scarcity vs. absolute scale. Figure 6 plots SAMBAL’s BLiMP advantage (pp) against the data-utilization measure of §4.5: the baseline’s accuracy as a percentage of the maximum SAMBAL accuracy at the same data budget. The 25 configurations span 8 data budgets (5%–100% of the corpus) and 4 model sizes (5M–30M parameters).

The relationship is strongly linear ($R^2 = 0.76$, Pearson $r = -0.87$), with a slope of about -0.47pp per percentage point of utilization: configurations where the baseline sits below $\sim 97\%$ of the SAMBAL ceiling consistently show $+1.5$ to $+3.2\text{pp}$ advantage, while those at or above the ceiling show none. Crucially, points from different data budgets intermingle rather than stratifying—e.g., 5M at 100% data (utilization $\approx 96\%$, $+1.8\text{pp}$) falls in the same region as 5M at 10% ($\approx 94\%$, $+2.6\text{pp}$) and 5M at 30% ($\approx 95\%$, $+3.2\text{pp}$), despite a $10\times$ difference in absolute token count. The advantage is thus predicted by how much headroom remains, not by which data budget produced it; by this analysis, a larger model trained on proportionally insufficient data would be predicted to show similar gains.

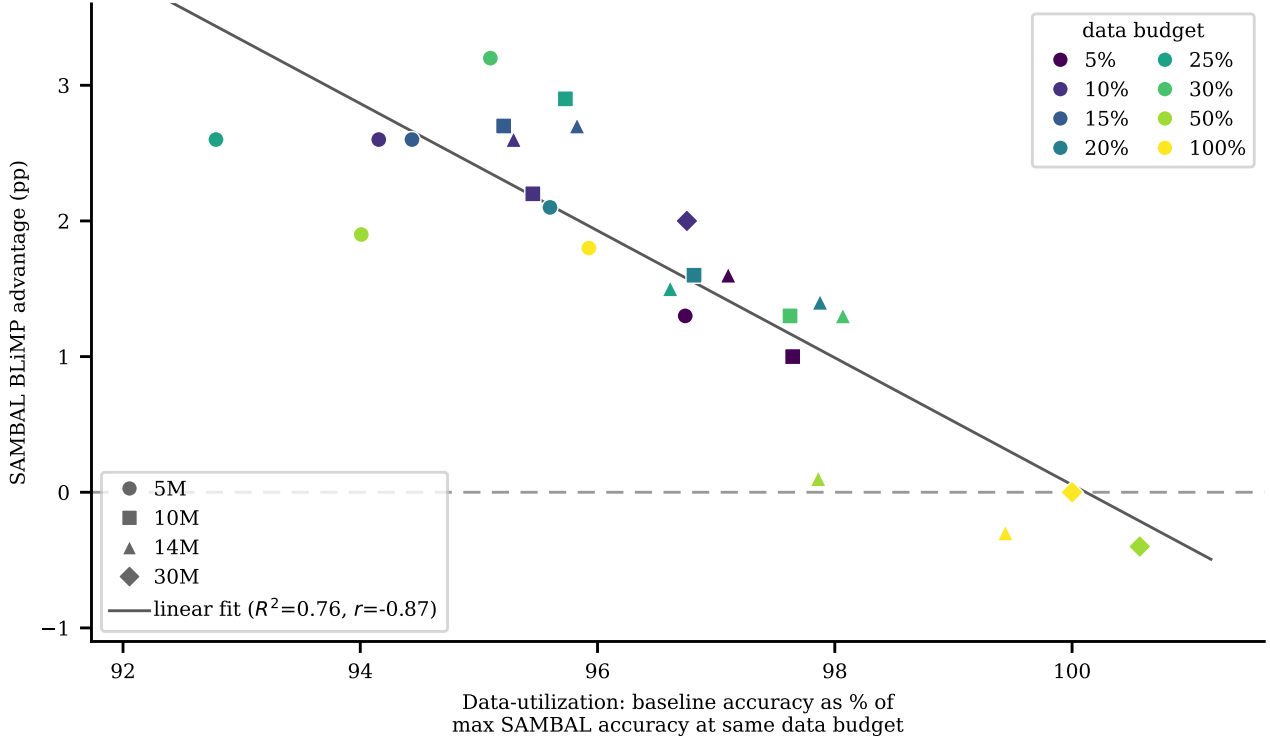


Figure 6. The efficiency advantage tracks relative resource scarcity, not absolute scale. SAMBAL’s BLiMP advantage (pp) vs. data-utilization (baseline accuracy as a percentage of the maximum SAMBAL accuracy at the same data budget), over 25 configurations (spanning 8 data budgets and 4 model sizes). Color encodes data budget and marker shape encodes model size. The advantage is strongly predicted by data-utilization ($R^2 = 0.76$, $r = -0.87$); points from different data budgets intermingle rather than separating by absolute token count.

H. Lexical-regularization control

SAMBAL narrows the lexical distribution as a side effect, so part of its BLiMP advantage could in principle stem from a simpler lexical distribution rather than from suppressing semantics. To isolate this effect while retaining meaning, we train a control on unmodified original text filtered to sentences containing only the same top-25k vocabulary SAMBAL uses, matching the 10% token budget of Fig. 4 (bottom). As Table 9 shows, the vocab-filtered control lands consistently above the baseline but below SAMBAL, so the simpler lexical distribution could explain part, but not all, of SAMBAL’s advantage.

Table 9. Lexical-regularization control (BLiMP %). A vocab-filtered baseline (original text restricted to SAMBAL’s top-25k vocabulary) at the 10% token budget improves over the baseline but does not reach SAMBAL.

Model size	baseline	Vocab-filtered	SAMBAL
5M	58.0	59.23 $\pm$ 0.71	60.6
14M	58.7	60.08 $\pm$ 0.29	61.3
30M	59.6	60.78 $\pm$ 0.76	61.6

Lexical entropy of the three corpora. We measure plug-in word-token n -gram entropies for each corpus (each with its own vocabulary and counts) at a matched budget of 4.90M word tokens, the size of the filtered corpus. Unigram entropy H_1 is effectively converged at this scale and is reported as a level; higher-order block entropies are not (plug-in estimates still rise with sample size), so we report conditional bigram entropy $h_2 = H_2 - H_1$ only as a same-budget difference from the baseline, where the shared downward bias cancels. Table 10 gives the numbers. The two interventions have distinct signatures: the vocabulary cut lowers H_1 by more than SAMBAL does while leaving h_2 essentially flat, whereas SAMBAL’s substitutions raise h_2 despite SAMBAL’s smaller vocabulary—local context becomes less predictive, as expected when co-occurrence structure is ablated.

Table 10. Lexical entropy at a matched budget of 4.90M word tokens (plug-in estimates, bits/word). Δh_2 is the change in conditional bigram entropy relative to the baseline at the same budget.

Corpus	H_1	Δh_2
baseline	10.80	—
SAMBAL	10.45	+0.57
Vocab-filtered (top-25k)	10.16	−0.04

We note that the three corpora also differ in effective vocabulary—baseline: 90,396, SAMBAL: 46,502, and vocab-filtered: 23,534 word types at the matched budget—which bears on the entropy values (SAMBAL’s count exceeds its 25k replacement vocabulary because that number does not include some of its replacement pools like proper names, and some rare words like digit strings pass through unablated, though this tail carries only $\approx 4\%$ of its tokens). Also, the BPE vocabulary the models train over is fixed at 8,192 regardless of corpus.

I. Additional LoTR generations

Prompt:	<i>'It's a dangerous business, Frodo, going out of your door' he used to say.</i>
SAMBAL	For Harrington, when I was little, there were three boxes of tubes on the guy, and they all came off as though they had sent some. And here I found 'Brazil,' this is what you know by Manipur Hungary. I'll be special with you. In this formation! We're not looking for any singing, glad? Cats have traveled solely over the basis that they'd been threatening and marveling at one another. Which is why we do it! Give me less.
baseline	You're getting out, but you need to take care of him and get him home again. Anyway, we should be hearing all about that. No! No! I'm not so sure. We can't do this without a frayed road. - Sorry, Lushi. They wanted the man who saw him and didn't know how to handle it all. But they knew that there was no bad way for him to get away. So, they went back into his house to see what happened. One day, Minnie met a little fox named Frodo who would fly from one corner to another
SAMBAL finetuned	'Then I hope you had better be off there.' He went on with his hand and drew himself up where the horse was clutching up Gollum along the slope. The two hobbits managed to go ahead, but they did not like it, even for a while; yet Frodo stared at them. 'Don't be absurd 'But you know more about it than I like, or this for certain.' He looked at Pippin and smiled aloud, but Sam could still see that the gleam in his eyes was not quite so easy as Frodo did; but it seemed and was becoming increasingly sick and at times so heavy that nothing else to do but follow him.
baseline finetuned	'I've been doing nothing wrong. I don't like it! ' Gollum fell down the flask and clashed into the water. Frodo raised his head and took Frodo back to his chair. He was silent for a moment, staring at him with an eyebrows lit by the fire. 'Don't let me sit 'I am not going out at all,' said Sam. 'You know what I mean. Go on and get off with you. But you won't forget it.' The old man sat down on the ground again and looked at him. 'What do you mean? What are you trespassing about in your face, Frodo?' asked Gand

Prompt:	<i>'The dark fire will not avail you, flame of Udun. Go back to the Shadow! You cannot pass,' bellowed Gandalf.</i>
SAMBAL	"Political danger!" bent the part to mark his present hands and work on the shirt. "I am fed up with my own scientists to discover an open deal of mud. A father at once made their contribution towards a world in the metal of his ants, but for some death it was not sod for them to Where is it? Equally available? It relies on the young study of Oakdale which has been called 'effectiveness'. but is hired by Australian, ... and rural McGlory Jura itself, in a world that it comes like an object of school. The happy family of Englishman and Navaj
baseline	"But I do think that is all this is done." The other man happily scampered towards him as he looked at himself in his eyes. But eldest hut was partly going, and riding down the road with Joshua to be brought up with a gripping cloak. "Good evening, big boys. Oh, "You are going after me!" exclaimed Tilly, her voice barely above the workbench. "I am going after you; I can't bear to let your brother leave." "But why? There will not be any time for a fray to go out until you start?" "Well, you must hurry up so soon," said
SAMBAL finetuned	'Then I must take your Ring,' he said. 'I am going to try and find Dain here beside him.' 'Yet you cannot!' cried Boromir; but they had been too sure that he would not reach the Shire for a while; yet he could see a look in his heart again, as if he were looking 'But we must sleep. We will lay on the boughs of Tirithlode at night.' They drew up their cloaks and turned far northward towards the hills, but they were not much troubled; for their threads writhed in a sheer scored behind, and then shrill mists quaved. Far away
baseline finetuned	'Then I have come into this place where there is no light!' He bowed and stood up. Sam ran round him with his head drawn back. 'You are coming quick, Mr. Frodo,' he said. 'Wait for me!' He went on. Pippin sat up and looked out. The sky was black as the night had 'I am not eager for a flight or two, but for some long web and endless search in which I must first find the lost house.' 'Then what do we do?' said Sam at last. Before he could speak they went to the banner with Gimli tucked his pony carved out of stone; then their hands slanted

J. PubMed specialization

As a robustness check that rapid adaptation is not specific to literary text, we repeat the LoRA fine-tuning experiment of §4.6 on a set of PubMed abstracts matched to the LoTR corpus size ($\approx 2,600$ abstracts). Table 11 shows the same trends as on LoTR: SAMBAL begins at much higher domain perplexity (255.4 vs. 40.5), yet after fine-tuning closes most of the gap (32.3) and substantially outperforms the best from-scratch model (58.54). Both models shed a few BLiMP points after specialization.

Table 11. Specialization on PubMed with LoRA fine-tuning.

Model	PubMed PPL	BLiMP Acc.
baseline (Long)	40.5	79.2
baseline (Long) fine-tuned	23.2	72.1
SAMBAL (Long)	255.4	79.3
SAMBAL (Long) fine-tuned	32.3	71.0
Best PubMed from-scratch model	58.54	55.74

K. Additional discussion: terminology and line drawing for SAMBAL

K.1. Syntax, semantics, grammar, and “world knowledge”

As in §3.1, we use *syntax* for the formal constraints that govern well-formedness (categories/features, dependencies, agreement, binding, complement structure) and *semantics* for meaning-bearing content (open-class lexical meaning, compositional interpretation, and background knowledge that makes an utterance informative or plausible). Within semantics we distinguish *world knowledge* (encyclopedic/factual and basic conceptual knowledge) from *semantic plausibility* (selectional and event-coherence cues); §4.3 probes the two separately, via EWoK and the swap probe respectively.

Two additional terminology details: First, we use *grammar* and *syntax* interchangeably. Second, we use *semantics* in the colloquial sense: it excludes the structural, compositional phenomena that formal semantics treats—quantifier scope, negation, binding, polarity licensing—because these are carried largely by closed-class items and configurational structure, which SAMBAL preserves (Table 1; cf. the SNLI discussion in §4.3). Under our usage, such phenomena pattern with grammar rather than with the semantic content we ablate.

In the paper we often compress this contrast into shorthand oppositions such as *grammar* vs. *plausibility* or *syntax* vs. *knowledge*. These shorthands are intentionally operational rather than precise: there exist meaning-like distinctions that are grammaticized and therefore directly affect acceptability. For SAMBAL, the practical question is whether a property must be preserved to avoid creating ungrammatical outputs (or flipping the label in a minimal pair), even if that property is not “pure syntax.”

K.2. Relation to the formal/functional competence distinction

Mahowald et al. (2024) distinguish *formal linguistic competence* from *functional linguistic competence* in evaluating LLMs. Their characterization of *formal* competence includes not only phonology/morphology and syntactic combination rules, but also *enough lexical meaning to know which words can go in which slots* and sensitivity to idiosyncratic constructions; *functional* competence involves using language in the world and draws on non-linguistic cognition (e.g., reasoning, world knowledge, situation tracking, social cognition). SAMBAL’s goal is narrower and more adversarial: we aim to keep the morphosyntactic scaffold that supports acceptability, while explicitly suppressing both encyclopedic facts and many of the low-level plausibility cues that ordinarily make utterances coherent and interpretable.

K.3. Why the line is hard to draw

The preserve/ablate boundary is inherently fuzzy because many semantic distinctions have grammatical reflexes:

- **Argument structure vs. plausibility:** verb meaning correlates with subcategorization. Preserving c-selection (e.g., whether a verb takes NP/PP/CP/to-inf) is grammatical; preserving “appropriate” thematic roles is plausibility.
- **Count/mass and determiner compatibility:** whether a noun behaves like a mass or count noun can determine whether a string with a given determiner is acceptable.
- **Animacy/humanness and binding/morphology:** pronoun/reflexive inventories encode properties that look semantic but are checked by the grammar.
- **Polarity sensitivity:** NPIs and the environments that license them are tightly linked to acceptability, even though they are also connected to meaning in the broad sense.
- **MWEs and constructional idiosyncrasy:** many MWEs behave like single lexical units; substituting within them often destroys grammatical well-formedness or yields parser-incompatible structures.

Because of these interactions, SAMBAL uses an explicitly operational criterion: if ablating a property reliably breaks well-formedness or flips acceptability in grammar-focused minimal pairs, we treat it as “grammar-relevant” for our purposes.

K.4. Borderline cases: gender/humanness and polarity sensitivity

Gender and humanness/animacy. Gender and humanness/animacy are borderline because they look like properties of referents, yet they are realized as morphosyntactic distinctions that directly affect acceptability (especially in pronominal/reflexive paradigms and binding). Ablating them would frequently create ungrammatical strings (or change which

member of a minimal pair is acceptable), rather than merely reducing topical interpretability. We therefore preserve these features by default and, when needed, constrain replacements so that these values remain stable.

This choice is also evaluation-driven: BLiMP contains minimal pairs that directly hinge on reflexive morphology (e.g., proper-name antecedents selecting *herself* vs. *himself*), and gender is entangled with other syntactic phenomena being tested (e.g., quantified antecedents and binding configurations). Changing gender/humanness would confound these tests.

NPIs and NPI licensors. Polarity sensitivity is similarly borderline: NPIs (e.g., *any*, *ever*) and the elements that license them (negation, downward-entailing quantifiers/determiners, and other polarity-conditioning material) can determine whether a sentence is acceptable. If SAMBAL were to freely relexicalize polarity items or their licensors, it could inadvertently flip grammaticality in NPI-focused minimal pairs and introduce “errors” that are not about syntactic well-formedness in the intended sense. Consequently, we preserve NPIs and NPI licensors as protected material in the default setting, treating them as grammar-critical guardrails even though they are meaning-adjacent.